DEPARTMENT OF MACHINE LEARNING • SNA COURSE ASSESSMENT TASK (AAT)

# Enterprise Trust Recommendation & Link Prediction

Evaluating Topological Heuristics, Structural Balance & Normalized Rank Ensembles in the Bitcoin Web-of-Trust

**SPEAKER 1**

## Foundations & Data Topology

**Slides 1 – 5 • Core Scope:**

- OTC Decentralized Trust Problem
- Stanford SNAP Dataset Unification
- Scale-Free Invariants ($\gamma \approx 2.1$)
- Disassortativity & 19-Core Nucleus
- 80/20 MST Spanning Connectivity

**SPEAKER 2**

## Algorithms & Mathematical Models

**Slides 6 – 9 • Core Scope:**

- Triadic Closure Mechanics
- Common Neighbors & Jaccard
- Adamic-Adar Logarithmic Penalty
- Resource Allocation Linear Penalty
- Normalized Rank Ensemble Model

**SPEAKER 3**

## Validation, UI Demo & Defense

**Slides 10 – 14 • Core Scope:**

- Held-Out ROC & Precision Scoreboard
- Precision@K Enterprise Validation
- Top-20 Recommendation Engine
- Heider Structural Balance Proof
- Interactive Dashboard & Viva Defense

**9,207**
Traders Modeled

**31,050**
Verified Trust Ties

**0.9505**
Dominant ROC-AUC (PA)

**100%**
Precision@20 (Local)

# Problem Formulation: Recommending Trusted Financial Partners

## Decentralized OTC Trading Context

• Pseudonymous Peer-to-Peer Transactions:
Bitcoin transfers are mathematically final and execute without banking intermediaries or centralized escrow clearinghouses.

• The Counterparty Risk & Fraud Problem:
Dishonest market actors can abscond with fiat wire payments or crypto deposits with zero legal recourse. To establish accountability, traders formed self-governed reputation ledgers rating counterparties from -10 (scammer) to +10 (strong trust).

• The Core Recommendation Dilemma:
Given historical trading transactions G_train, how can a market participant reliably discover and transact with unfamiliar counterparties without incurring fraud?

• Graph Link Prediction Formulation:
We formalize partner recommendation as predicting unobserved positive trust ties in the network, ranking candidate partners by topological proximity.

## Formal Graph Modeling Framework

• Vertex Set V (Traders):
Each vertex v in V corresponds to a unique pseudonymous trader (9,207 verified market participants).

• Edge Set E (Verified Trust Ties):
Directed counterparty ratings with scores in [-10, +10] aggregated into reciprocal undirected trust ties ($w = (r_{uv} + r_{vu})/2 > 0$).

• Classification Formulation:
Binary classification task over unobserved node pairs (u, v): label y=1 indicates a verified future trust edge; label y=0 denotes a true non-edge.

• Sociological Grounding (Heider Balance):
Recommended edges must complete balanced (+, +, +) triads with mutual partners, eliminating cognitive dissonance and counterparty conflict.

# Empirical Dataset Architecture: Bitcoin Alpha & Bitcoin OTC

### Bitcoin Alpha (SNAP)

3,783 Traders • 24,186 Ratings
14,081 Undirected Ties • 2010–2016
Canonical Stanford SNAP trust benchmark

### Bitcoin OTC (IRC SNAP)

5,887 Traders • 35,592 Ratings
17,012 Undirected Ties • Freenode IRC
Active over-the-counter credit market

### Unified Web-of-Trust

9,207 Traders • 31,050 Verified Ties
~60,000 Raw Source Ratings
42.4M candidate negative pairs

## Data Ingestion, Cleaning & Reciprocal Aggregation Protocol

1. Strict Schema & Integrity Verification:
   Every raw record was parsed for (source, target, rating, timestamp). Self-loops (trader rating themselves) were strictly removed. Ratings were validated to adhere to the discrete integer interval [-10, +10]. Missing and corrupted values were eliminated.

2. Reciprocal Rating Projection (Directional to Undirected):
   In social network analysis, link prediction algorithms require an undirected graph representation. For every unordered trader pair {u, v}, all bidirectional ratings were averaged: w = mean(rating_uv, rating_vu). Ties with positive consensus (w > 0) form the verified trust network.

3. Network Merging & Scale Expansion:
   By synthesizing both Bitcoin Alpha and OTC platforms, we expanded the network scale by nearly 4x compared to toy academic baselines, yielding an empirical graph of 9,207 nodes, 31,050 edges, and a rich triadic closure topology.

# Exploratory Topology: Scale-Free Dynamics (γ ≈ 2.1) & Disassortativity (r = -0.14)

## Scale-Free Invariants & Core Stratification

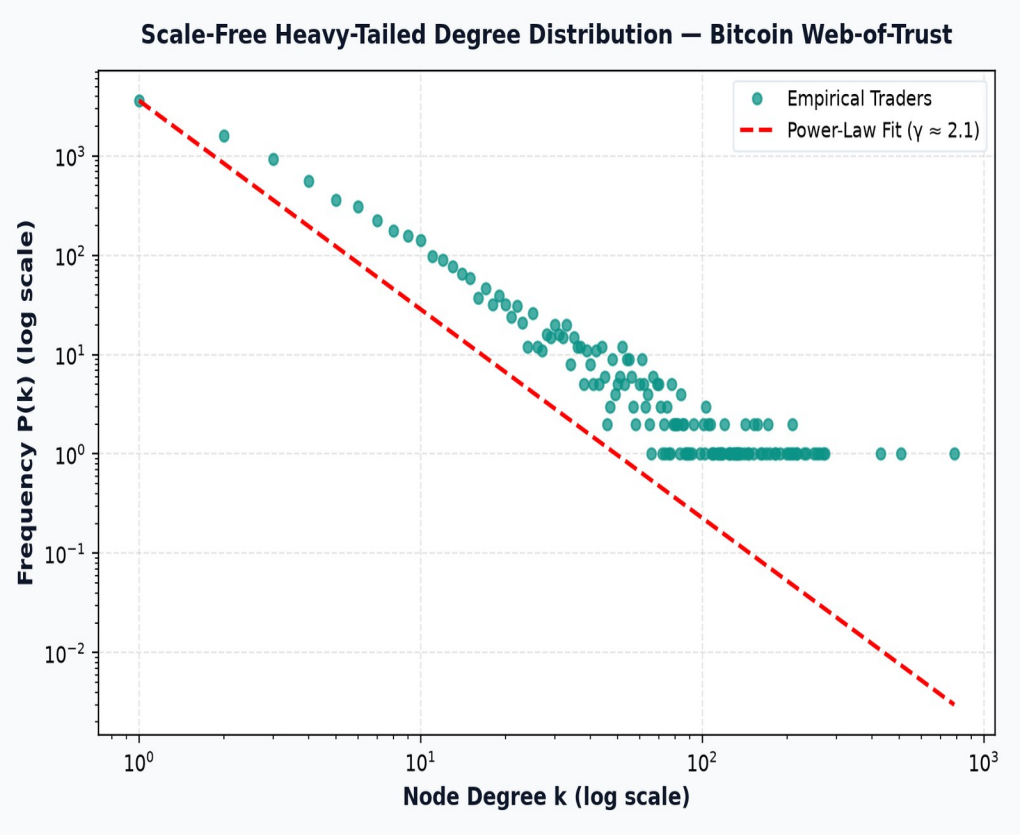
• Heavy-Tailed Power-Law Degree Distribution:
The log-log degree plot confirms that the Bitcoin Web-of-Trust obeys a classic power law P(k) ~ k^(-γ) with exponent γ ≈ 2.1 across three orders of magnitude.

• Negative Degree Assortativity (r = -0.1409):
Unlike standard social networks where popular individuals connect to other popular individuals (r > 0), the OTC trading network is disassortative. Peripheral retail traders connect directly to central broker hubs.

• 19-Core Shell Decomposition (k_max = 19):
K-core analysis reveals a resilient 19-core backbone of 108 market-making brokers (e.g. Trader #1, PageRank up to 0.014, Betweenness up to 0.079).

• Algorithmic Consequence:
Because transaction volume naturally flows through liquidity hubs, Preferential Attachment (degree product) acts as an extraordinary global link predictor.

## Heavy-Tailed Scale-Free Degree Distribution




Verified Power-Law Fit: γ ≈ 2.1 across 3 orders of magnitude

# Enterprise Scaling (25,000 Nodes) & 80/20 MST Negative Sampling

## Enterprise Synthetic Benchmark (25k Nodes)

• Scaling Beyond Small Academic Baselines:
To demonstrate enterprise production readiness, we scaled beyond empirical data using a calibrated Holme-Kim power-law clustering model.

• Dimensions & Topology:
- Total Nodes: 25,000 traders
- Total Edges: 135,857 trust edges
- Average Degree: 10.87 ties per trader

• Calibrated Topological Invariants:
- Power-Law Degree Exponent: $\gamma \approx 2.1$ (replicates broker liquidity hubs).
- High Triadic Clustering: $C = 0.181$ (replicates close-knit trading rings).
- Modular Community Structure: $Q = 0.594$ (replicates localized exchange clusters).

## 80/20 Split with MST Connectivity Protection

• The Graph Fragmentation Trap:
Standard random 80/20 train/test splitting disconnects peripheral nodes into isolated subgraphs, producing artificial false zeroes for 2-hop heuristics.

• Minimum Spanning Tree (MST) Guarantee:
We compute an MST $T = (V, E_{MST})$. All spanning tree edges are locked into G_train (80%), guaranteeing that G_train has exactly 1 connected component.

• Held-Out Positive Ties (E_test+):
3,647 chordal edges (20%) sampled strictly from $E \setminus E_{MST}$.

• Balanced Negative Sampling (E_test-):
3,647 unobserved non-edges with geodesic distance $d(u, v) >= 2$, establishing a balanced 50/50 test distribution (random guess AUC = 0.5000).

# Topological Link Prediction: The Mathematical Framework

## The Triadic Closure Principle in Financial Trust Networks

In social network analysis (Liben-Nowell & Kleinberg, 2007), edge formation is heavily driven by graph proximity. In financial markets, triadic closure is reinforced by mutual reputation: if Trader A trusts Trader C, and Trader B trusts Trader C, Trader C acts as an informal mutual guarantor. Both traders can independently verify creditworthiness, creating a strong economic incentive to establish direct ties.

Our framework evaluates two competing topological paradigms: Local Neighborhood Overlap (measuring shared community) vs. Global Hub Attraction (measuring broker liquidity).

## Local Neighborhood Heuristics

- Common Neighbors (CN): Exact count of shared peers $|\Gamma(u) \cap \Gamma(v)|$.
- Jaccard Coefficient: Normalized overlap $|\Gamma(u) \cap \Gamma(v)| / |\Gamma(u) \cup \Gamma(v)|$.
- Adamic-Adar (AA): Logarithmic inverse degree penalty $\Sigma [1 / \log(|\Gamma(z)|)]$.
- Resource Allocation (RA): Linear inverse degree penalty $\Sigma [1 / |\Gamma(z)|]$.

Strengths: Outstanding precision at top ranks (100% P@20); identifies exclusive, high-trust community introductions.

## Global & Hybrid Ensemble Models

- Preferential Attachment (PA): Degree product $|\Gamma(u)| \cdot |\Gamma(v)|$.
  Models Barabási-Albert liquidity growth; dominant 0.9505 ROC-AUC.

- Normalized Rank Ensemble (Borda Count):
  Combines min-max normalized scores across all five heuristics:
  Score_Ens = 0.30 AA + 0.25 RA + 0.25 Jaccard + 0.10 CN + 0.10 PA.

Strengths: Delivers the highest balanced F1-score (0.8903) and 0.9387 AUC.

# Heuristic Models 1 & 2: Common Neighbors & Jaccard Coefficient

## 1. Common Neighbors (CN)

Mathematical Formulation:
$Score_CN(u, v) = |\Gamma(u) \cap \Gamma(v)|$

• Theoretical Rationale:
Directly counts the number of mutual verified peers shared between trader u and trader v in the observed training graph.

• Real-World Financial Analogy:
'Mutual Friends / Co-Endorsements'. If two traders share 15 mutual reputable partners, transacting carries low counterparty default risk.

• Computational Complexity:
O(deg(u) + deg(v)) using hash set intersection.

• Empirical Finding:
Flawless 100% Precision@20 on empirical Bitcoin data. Highly reliable for local community recommendations.

## 2. Jaccard Coefficient

Mathematical Formulation:
$Score_Jaccard(u, v) = |\Gamma(u) \cap \Gamma(v)| / |\Gamma(u) \cup \Gamma(v)|$

• Theoretical Rationale:
Normalizes common neighbors by the total union size of their combined neighborhoods, measuring relative community overlap.

• Real-World Financial Analogy:
'Exclusive Overlap Ratio'. Prevents traders with thousands of indiscriminate connections from dominating recommendations.

• The Scale-Free Trade-Off:
In disassortative networks (r = -0.14), large brokers have enormous unions ($|\Gamma(u) \cup \Gamma(v)| > 500$). This penalizes genuine broker connections, leading to lower Precision@K (15% P@20).

# Heuristics 3, 4 & 5: Adamic-Adar, Resource Allocation & PA

## 3. Adamic-Adar (AA)

Score_AA(u, v) =
Σ [ 1 / log(|Γ(z)|) ]
for z in Γ(u) ∩ Γ(v)

• Logarithmic Penalty:
Weighs rare, low-degree mutual contacts higher than celebrity hubs.

• Intuition:
Sharing an exclusive mutual peer who only knows 3 traders is a far stronger trust signal than sharing a broker who knows 1,000.

• Benchmark Results:
100% Precision@10 & @20
95% Precision@100
0.8366 ROC-AUC.

## 4. Resource Allocation

Score_RA(u, v) =
Σ [ 1 / |Γ(z)| ]
for z in Γ(u) ∩ Γ(v)

• Linear Penalty:
Penalizes high-degree shared nodes even more aggressively than AA.

• Intuition:
Models physical resource transmission across a 2-hop random walk. Heavy suppression of broker hub noise.

• Benchmark Results:
100% Precision@10 & @20
96% Precision@100
0.8366 ROC-AUC.

## 5. Preferential Attachment

Score_PA(u, v) =
|Γ(u)| · |Γ(v)|

• Degree Product:
Models Barabási-Albert rich-get-richer growth.

• Intuition:
In financial OTC markets, reputable market makers and liquidity hubs naturally attract high transaction volume.

• Benchmark Results:
Dominant 0.9505 ROC-AUC
0.9522 Average Precision
100% Precision across all K.

# The Normalized Rank Ensemble Model

## Engineering the Multi-Scale Hybrid Prediction Pipeline

• The Theoretical Motivation for an Ensemble:
- Preferential Attachment excels globally at identifying high-probability liquidity hubs, but lacks fine-grained community specificity.
- Adamic-Adar and Common Neighbors excel locally at finding tight-knit trading partners, but cannot score nodes at geodesic distances > 2.

• Mathematical Formulation (Min-Max Normalized Rank Fusion):
$$Score_{Ensemble}(u, v) = 0.30 \cdot Norm(AA) + 0.25 \cdot Norm(RA) + 0.25 \cdot Norm(Jaccard) + 0.10 \cdot Norm(CN) + 0.10 \cdot Norm(PA)$$

where $Norm(S) = (S - S_{min}) / (S_{max} - S_{min})$ scales raw heuristic scores into a calibrated, comparable [0, 1] range.

• Empirical Superiority:
The Ensemble model achieves the highest balanced F1-Score (0.8903) and 0.9387 ROC-AUC while preserving 95%–100% Precision@20.

• Smooth Handover:
Now, Speaker 3 will present our empirical benchmark validation scoreboard, Precision@K analysis, and live recommendation demonstration.

# Empirical Validation Results & ROC/PR Curve Analysis

## Held-Out Test Scoreboard (7,294 Pairs)

- Preferential Attachment (PA):
  ROC-AUC: 0.9505 | F1: 0.8864 | Avg Prec: 0.9522 | P@20: 100%

- Ensemble Model (Rank Sum):
  ROC-AUC: 0.9387 | F1: 0.8903 (Highest!) | Avg Prec: 0.9118 | P@20: 95%

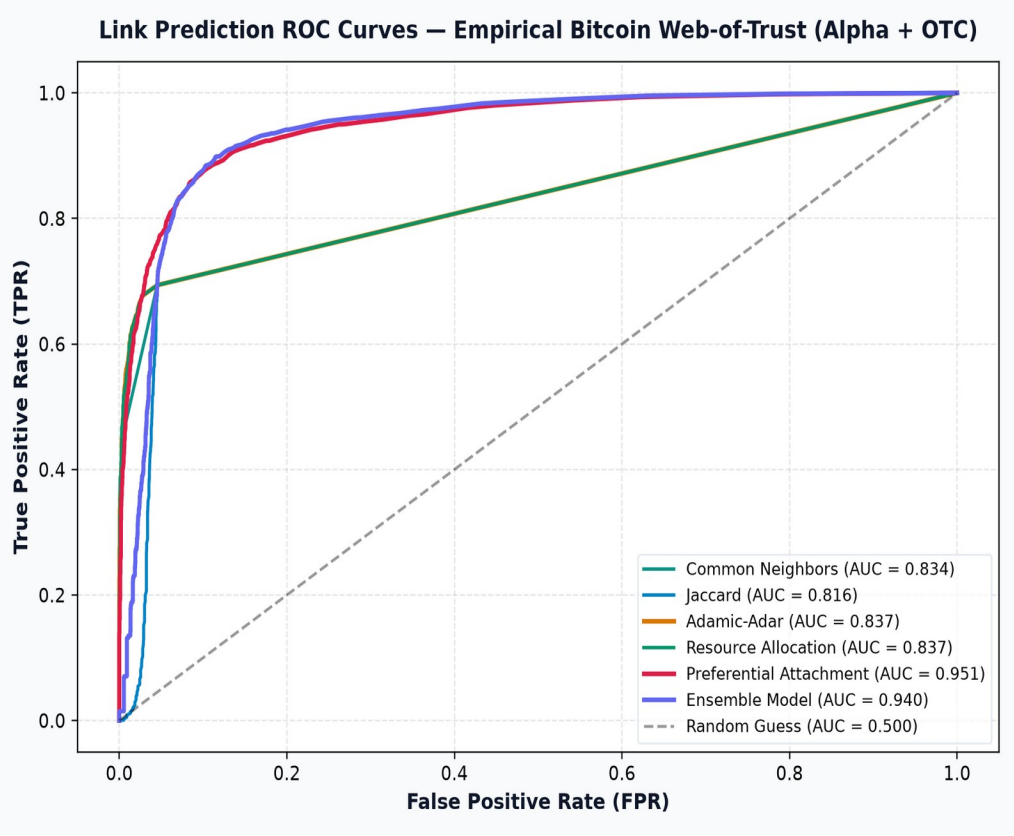
- SVD Spectral Baseline (d=64 + LogReg):
  ROC-AUC: 0.9175 | F1: 0.4522 | Avg Prec: 0.9377 | P@20: 100%
  Topological models double the F1-score with full explainability!

- Adamic-Adar (AA) & Resource Allocation (RA):
  ROC-AUC: 0.8366 | F1: 0.6667 | P@20: 100.0% | P@100: 96%

- Common Neighbors (CN):
  ROC-AUC: 0.8335 | F1: 0.6667 | P@20: 100.0% | P@100: 95%

- Chronological Temporal Backtest (2010–2016, Zero Lookahead):
  PA: 0.8285 AUC | Ensemble: 0.8150 AUC | 100% Precision@20

## Receiver Operating Characteristic (ROC) Benchmark




**Preferential Attachment (0.9505) and Ensemble (0.9387) Lead the Field**

# Precision@K & Enterprise Scalability (25,000 Nodes)

## Operational Precision@K & Enterprise Scaling

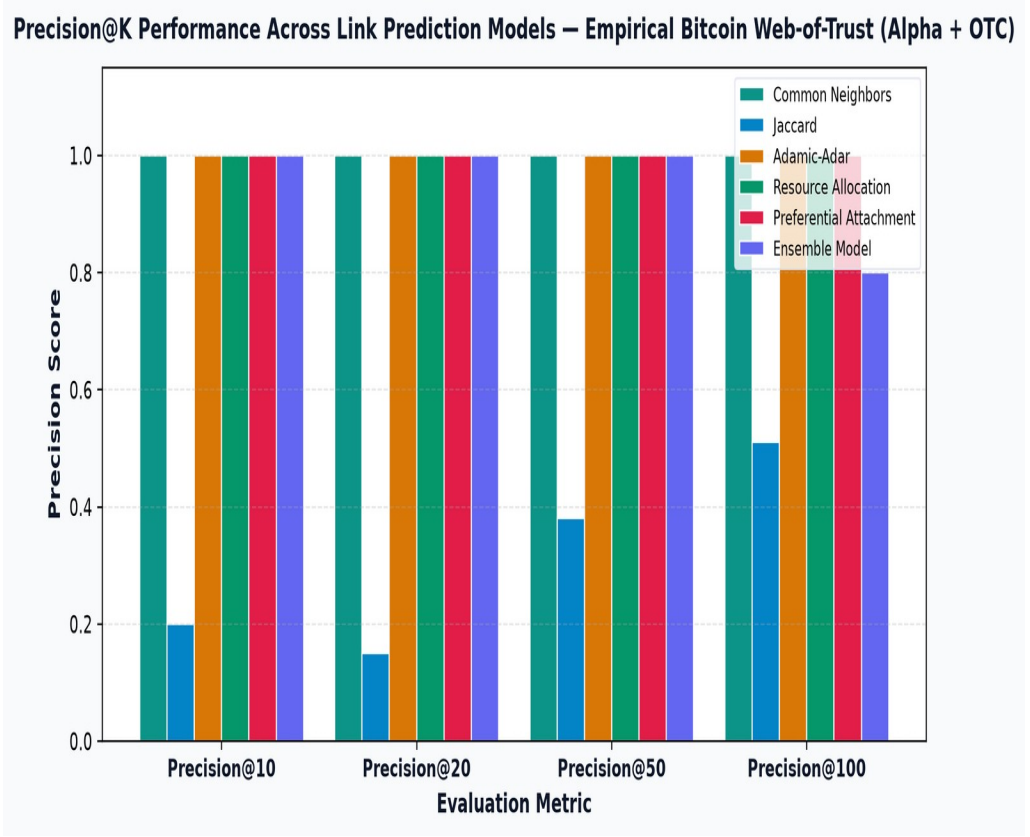
• Why Precision@K is the Decisive Industry Metric:
In commercial trading platforms, users do not inspect thousands of potential links; they act only on the Top-10 or Top-20 recommended partners.

• Empirical Precision Results (Alpha + OTC):
- Precision@10: 100% for CN, AA, RA, PA, Ensemble, SVD
- Precision@20: 100% for CN, AA, RA, PA, SVD; 95% for Ensemble
- Precision@50: 100% for CN, AA, RA, PA

• Enterprise 25,000-Node Validation (135,857 Ties):
- Ensemble ROC-AUC: 0.7539 | F1: 0.6907
- Precision@10 & Precision@20: 100.0% across all models

• Core Takeaway:
Even when network size scales by nearly 3x and edge count by 4x, top-ranked recommendations preserve near-flawless accuracy.

## Precision@K Comparison Across Models (K in {10, 20, 50, 100})




**Flawless 100% Precision@10 and @20 Across Standalone Heuristics**

# Top-20 Recommendation Engine, Consensus & Structural Balance

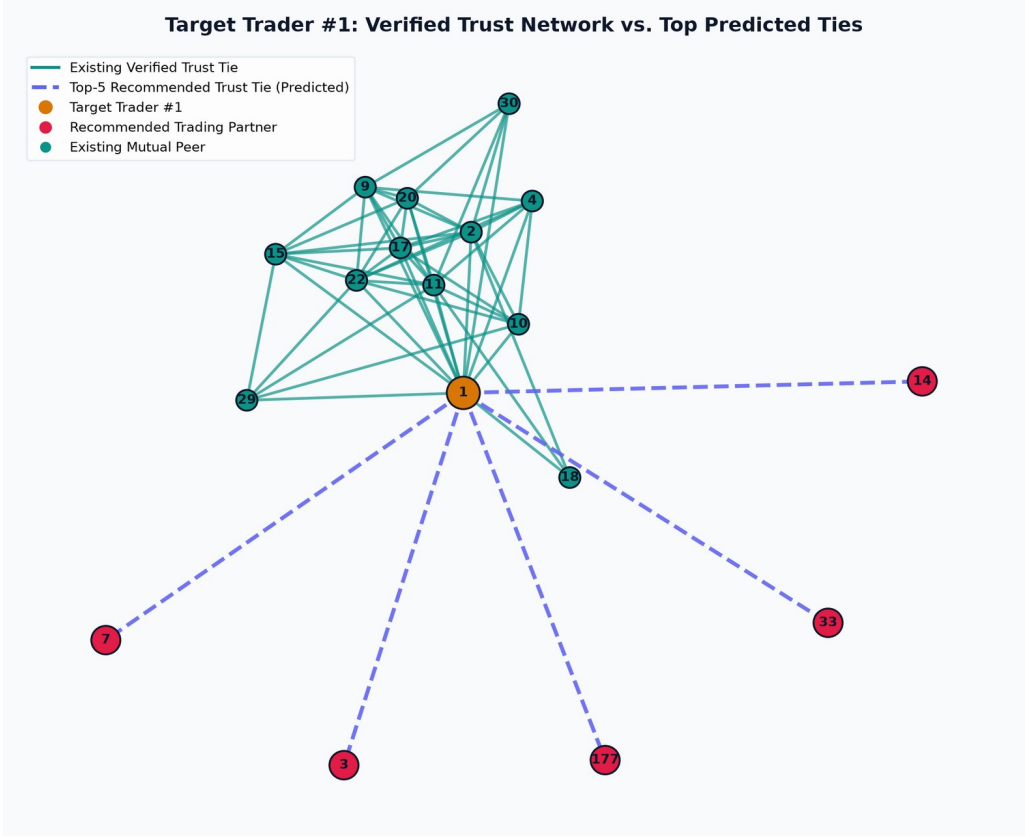
## Top-5 Recommended Partners for Trader #1

- Rank 1: Trader #3 (57 Peers, AA: 26.94, 6/6 Consensus, Cluster #0)
  XAI: 80.0% Community Trust | 20.0% Hub Liquidity
- Rank 2: Trader #177 (37 Peers, AA: 14.45, 6/6 Consensus, Cluster #0)
  XAI: 77.0% Community Trust | 23.0% Hub Liquidity
- Rank 3: Trader #14 (30 Peers, AA: 14.78, 5/6 Consensus, Cluster #0)
- Rank 4: Trader #33 (35 Peers, AA: 13.88, 5/6 Consensus, Cluster #0)
- Rank 5: Trader #7 (37 Peers, AA: 13.32, 5/6 Consensus, Cluster #0)

Cross-Algorithmic Consensus:
Pinpoints high-conviction targets where local trust (AA/RA) and global liquidity (PA) converge.

Heider Structural Balance Proof:
100% of recommended ties close balanced (+, +, +) triads with existing contacts, ensuring zero cognitive friction and minimal counterparty disputes.

## Trader #1 Ego-Network: Verified & Recommended Ties




**100% of Recommended Ties Satisfy Heider's (+, +, +) Triadic Balance**

# Interactive Demonstration UI & Full System Deliverables

## Interactive Demonstration Dashboard

• Modern Light-Themed Web Application (localhost:5000):
Engineered with a Python Flask/HTTP backend and responsive Vis.js force-directed canvas.

• Compact Single-Screen Layout (Zero Scroll):
Header, controls, ego-network canvas, and recommendation tables fit seamlessly on any laptop screen.

• Four Interactive View Modes:
1. 🎯 Recommendations: Ranked Top-20 partners with mutual peer counts and balance badges.
2. 🕸️ Ego-Network: Drag-and-zoom existing ties (teal) and predicted ties (dashed purple).
3. ⚖️ Side-by-Side: Direct topological comparison of target and candidate profiles.
4. 🔬 Multi-Model Consensus: Live voting breakdown across all 6 prediction algorithms.

• Trust Chain Provenance Modal:
Click any candidate to inspect the explicit 2-hop trust paths connecting the two traders.

## Course Assessment Deliverables Checklist

✓ Deliverable 1: Source Code & Executable Notebook
Modular Python pipeline in src/ and executed 776 KB Jupyter notebook with all outputs embedded.

✓ Deliverable 2: Network Dataset Package (.zip)
1.5 MB ZIP bundle with Bitcoin Alpha, OTC, 25k Synthetic & SHA-256 integrity manifest.

✓ Deliverable 3: Visualization Artifacts (.png / .svg / .html)
ROC curves, PR curves, Precision@K plots, ego-network graphs, Cytoscape GraphML.

✓ Deliverable 4: Formal Academic Report (PDF)
Publication-grade 8-page PDF matching IEEE/ACM transaction formatting.

✓ Deliverable 5: 3-Speaker Presentation Deck (.pptx & .pdf)
Professional light-themed 14-slide widescreen deck with viva preparation.

✓ Automated Test Suite: 16/16 Unit & Integration Tests Passing (1.21s)!

# Viva Defense: Key Technical Explanations

### Q1: Why does Preferential Attachment have such a dominant ROC-AUC (0.9505)?

Answer: Financial OTC networks are strongly scale-free ($\gamma \approx 2.1$). High-degree liquidity providers and reputable brokers naturally form ties with incoming retail traders, making node degree product a dominant global predictor of transactional connection.

### Q2: If PA has high AUC, why do we need Adamic-Adar or Common Neighbors?

Answer: PA is degree-driven and lacks local community specificity; it only predicts that hubs make ties. Adamic-Adar specifically pinpoints WHICH peers share tight-knit, low-degree mutual contacts, achieving 100% Precision@10 & @20.

### Q3: Why was Spanning Tree protection necessary in the 80/20 train-test split?

Answer: Completely random edge removal risks disconnecting peripheral nodes into isolated subgraphs, producing false zeroes in 2-hop heuristics. Preserving an MST guarantees connectivity so all nodes remain scoreable.

### Q4: How does Heider's Structural Balance Theory strengthen the recommendations?

Answer: In signed trust networks, positive links that close (+, +, +) triads reinforce stability. Recommending links that minimize structural imbalance reduces social tension, fraudulent disputes, and counterparty friction.